

TUTOR-Bench: A Target-Conditioned Utility Benchmark for Cross-Task Robot Data Reuse

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Abstract

Robot demonstrations are costly because each episode requires expert or scripted execution. Cross-task reuse could improve the use of existing corpora, but under a fixed training budget, donor samples displace target exposure and can help or harm completion. Measuring a candidate donor's effect requires retraining the target-donor pair and running closed-loop rollouts; metadata and semantic similarity provide proxies rather than outcomes. Exhaustive validation is therefore costly. We introduce TUTOR-BENCH, a directed outcome dataset for this decision. A prespecified target-only screen selects 14 tasks from four 10-task LIBERO suites before mixed outcomes are observed. For all 182 ordered target-donor pairs, a target-specific Diffusion Policy receives 90% target and 10% donor exposure under a fixed update budget and is compared with target-only training at the same checkpoint on ten matched scenarios. Their completion-rate difference defines signed utility, turning each intervention into a reusable outcome label. TUTOR-BENCH supports role-disjoint *Donor Selection for New Targets* and *Budgeted Utility Completion* with limited outcomes. Among 182 cells, 40 are positive, 12 are zero, and 130 are negative; pair-weighted mixed success is 31.7% versus 46.7% without a donor, yielding mean utility of -15.0 percentage points (pp). Role-disjoint evaluation establishes NoDonor as a competitive safety baseline and identifies the positive cells as concrete targets for improved selectors. With five measured cells, Decision Utility rises from 0.0 to 5.8 pp; diversity-first querying also lowers donor-decision regret relative to random acquisition, while its full-row reconstruction advantage remains unresolved. Together, TUTOR-BENCH turns costly pairwise interventions into a reusable, auditable outcome reference for developing donor selectors and measurement-efficient acquisition strategies. Code and benchmark resources are publicly available at <https://github.com/Roswellii/TUTOR-Bench>.

CCS Concepts

• Information systems → Data mining; • Computing methodologies → Learning from demonstrations; Reinforcement learning.

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Keywords

robot learning, cross-task data reuse, dataset benchmark, data valuation, negative transfer, imitation learning

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1 Introduction

Modern robot policies increasingly learn from demonstrations pooled across many tasks [8, 12, 13]. A vision-language-action (VLA) policy maps visual observations and a language instruction to robot actions, while a demonstration episode records time-ordered observations, robot states, and expert actions. Policy quality is ultimately judged in a *closed-loop rollout*, where each action changes what the policy observes next. When training and evaluating a policy for one *target task*, a practitioner must therefore decide whether demonstrations from another task—a *donor task*—should supply part of the training samples. Under a fixed learner, target episode pool, and update budget, which donor, if any, should be used? Donor experience may provide useful visual or manipulation structure, while donor draws displace target-task exposure. Donor choice is therefore a budgeted allocation decision, not simply a question of adding more data.

Making this choice before training is difficult. Similar instructions can hide different geometry or action sequences, whereas different instructions can share reaching, grasping, or placement structure. Data-selection, influence, valuation, and task-affinity methods estimate such relationships from language and visual embeddings, trajectories, gradients, or surrogate losses [1, 3, 9, 10]. Viewed as a data-mining problem, tasks are entities, an ordered task pair has a costly outcome label, and pre-intervention task and trajectory descriptors are features for predicting that label. Proxy scores make prediction affordable; evaluating them requires a common outcome reference that records how target completion changes after training with each candidate donor. Robot corpora provide expert trajectories, while selection studies

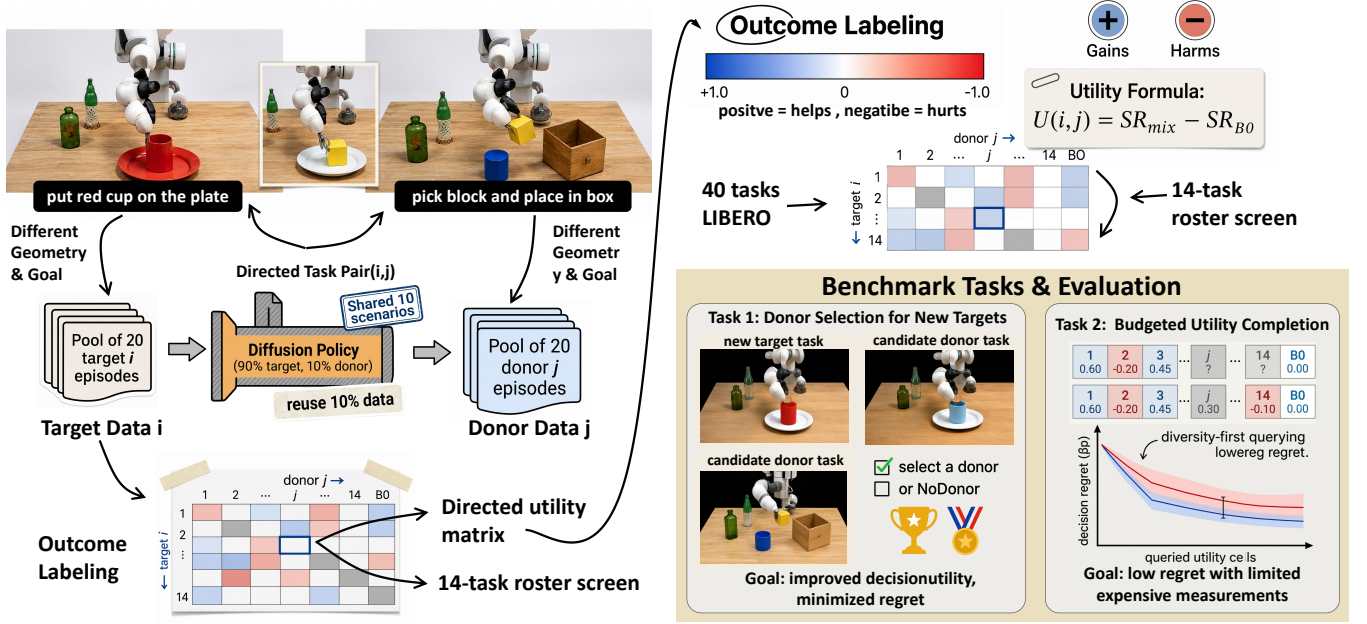


Figure 1: Overview of TUTOR-Bench construction and its two benchmark tasks.

usually evaluate only the chosen subsets. A complete table of matched train-and-evaluate outcomes under one learner, mixture ratio, checkpoint rule, and scenario set supplies the shared answer key needed to compare donor-selection methods.

We call this resource TUTOR-BENCH, with TUTOR denoting *Target-conditioned Utility for Task-Oriented Reuse*. The name emphasizes that utility is defined relative to a particular target and operational reuse decision, rather than as an intrinsic similarity or value of the donor data. Figure 1 summarizes the resulting outcome construction and benchmark interfaces.

We construct TUTOR-BENCH from the 40 tasks in four LIBERO tabletop- manipulation suites [11]. Before observing any mixed-policy outcome, an audited target-only screen retains the 14 tasks that clear a prespecified success threshold, limiting floor effects. For every ordered pair among these tasks, a target-specific Diffusion Policy learner [2] is trained under matched controls. The mixed condition keeps all 20 target episodes available but allocates 10% of training draws to the donor; the target-only condition draws only from the target. Both use the same learner, update budget, target-specific checkpoint, and evaluation scenarios. The mixed-minus-target-only difference in task-completion rate defines signed, directed utility, producing $14 \times 13 = 182$ measured target-donor outcomes. Selecting NoDonor is an explicit action with utility zero. The label generator is a target-specific Diffusion Policy, while language, images, and trajectories remain available as outcome-free inputs for methods motivated by VLA data reuse.

The benchmark separates available inputs, hidden outcomes, and deterministic scoring. Its primary task, *Donor*

Selection for New Targets, asks a method to score 13 donors or select NoDonor for a target identity excluded from all fitted utility labels. Fourteen role-disjoint leave-one-target-out folds score each complete target row once. The secondary task, *Budgeted Utility Completion*, reveals a limited number of costly pair outcomes and evaluates both recovery of the remaining utilities and the resulting donor decision.

The completed matrix contains 40 positive, 12 zero, and 130 negative cells. Pair-weighted mixed success is 31.7%, compared with 46.7% for target-only training, for a mean utility of -15.0 percentage points (pp). Utilities range from -63.3 to $+26.7$ pp, and the mean absolute reciprocal difference is 22.3 pp, showing that aggregate harm coexists with directed, target-dependent gains. Role-disjoint evaluation establishes NoDonor as a competitive safety baseline and identifies the 40 positive cells as concrete targets for improved selectors. In Budgeted Utility Completion, five measurements raise Decision Utility from 0.0 to 5.8 pp, and diversity-first querying lowers donor-decision regret relative to random acquisition; the NFE AUC difference remains unresolved. Together, these findings support both conservative abstention and measurement-efficient acquisition, while providing a controlled reference for future learners, mixture ratios, embodiments, and VLA models.

Our contributions are threefold:

- (1) **A directed outcome dataset for cross-task robot data reuse.** We register 182 ordered target-donor outcomes among 14 selected LIBERO tasks, providing measured train-and-evaluate consequences for developing and comparing reuse signals.
- (2) **A controlled and auditable construction protocol.** A fixed learner, 9:1 exposure rule, target-specific

checkpoint, and matched rollout scenarios link every utility to its episodes, training runs, and evaluation records.

(3) Decision- and cost-aware benchmark interfaces.

Role-disjoint Donor Selection for New Targets evaluates donor choice with explicit abstention, while Budgeted Utility Completion evaluates recovery and decisions after limited measured interventions.

2 Related Work

The growing scale and diversity of robot-learning corpora have made cross-task reuse a central route to more general policies. Open X-Embodiment aggregates heterogeneous trajectories across robots and domains, and Octo and OpenVLA use such broad supervision to train generalist policies [8, 12, 13]. Complementing these large-scale resources, LIBERO evaluates knowledge transfer across manipulation tasks [11]. This line of work establishes that experience can be shared across tasks, but the availability of a large corpus does not determine which part of that corpus should be reused for a particular target. Recent robot-data curation methods address this choice more directly: DataMIL estimates demonstration value through datamodels, whereas CUPID and *Quality over Quantity* use influence-based signals to rank or filter demonstrations [1, 3, 9]. Their outputs are selected subsets or predictive scores; they do not provide a common reference of pairwise closed-loop outcomes under matched training interventions.

This distinction connects robot-data curation to the broader literature on data valuation and transferability. Data Shapley assigns value through marginal contributions, while EcoVal seeks a more efficient approximation for examples or data groups [5, 17]. At the task level, Grad-TAG estimates gradient-based affinities, DoReMi learns mixture weights, and PCGrad modifies conflicting gradients during multitask training [10, 19, 20]. These approaches avoid the cost of exhaustively retraining every candidate combination and are useful for selection or optimization. A proxy affinity or marginal value, however, need not equal the realized effect of assigning a specified donor to a specified target. The outcome studied in TUTOR-BENCH is therefore directional and intervention-specific: it records the signed change from matched target-only training while holding the learner and update budget fixed.

Reliable comparison of such outcomes also depends on dataset and benchmark design. Datasheets formalize reporting of composition, provenance, intended use, and limitations [4]; SimCoachCorpus links multimodal embodied trajectories to task-relevant performance measurements [16]. Work on budgeted dataset selection further shows that a limited set of evaluations can be organized to preserve downstream model comparisons [6]. TUTOR-BENCH brings these ideas together at the donor–target level. Each utility is traceable to source episodes, training records, and paired rollouts, while one outcome reference supports both Donor Selection for New Targets and Budgeted Utility Completion. Its contribution

is thus an auditable intervention–outcome layer for evaluating reuse decisions, rather than another donor-selection algorithm.

3 TUTOR-Bench Dataset

TUTOR-BENCH connects source demonstration episodes, controlled target–donor interventions, matched rollouts, utility labels, and audit provenance. Its construction protocol fixes the task registry, target-only roster screen, training intervention, and matched utility estimator before donor outcomes are used. Characterization then examines roster coverage, directional utility structure, seed sensitivity, semantic validity, and artifact integrity.

3.1 Dataset Construction

Each record is an ordered target–donor intervention (i, j) linking source demonstration episodes, target-only and mixed runs, matched target rollouts, a utility label, and provenance. Reversing the order changes the optimized task, target data, and baseline, so (i, j) and (j, i) are distinct. The source registry starts from all 40 tasks in the four ten-task LIBERO suites [11]; a deterministic manifest selects 20 demonstration episodes per task, yielding 800 globally unique episodes and 133,328 recorded steps. All candidates remain in the check-summed provenance registry, including those that do not enter the benchmark roster or intervention matrix.

The target-only screen uses no mixed outcome. For target i , training seed s , checkpoint t , and scenario e , let $z_{i,s,t,e}^0 \in \{0, 1\}$ denote target-only success. On the ten-scenario calibration set $\mathcal{E}_{\text{cal}}$ and fixed seed s_{cal} , the task-specific checkpoint is

$$t_i^* = \underset{t \in \{5,000, 6,000, \dots, 10,000\}}{\operatorname{argmin}} \left| \frac{1}{|\mathcal{E}_{\text{cal}}|} \sum_{e \in \mathcal{E}_{\text{cal}}} z_{i,s_{\text{cal}},t,e}^0 - 0.5 \right|, \quad (1)$$

with earlier ties. This midpoint is an operating point with room to observe both gains and harms, not an estimate of final convergence. After freezing t_i^* , write $z_{i,s,e}^0 := z_{i,s,t_i^*,e}^0$. A separate target-only evaluation uses the three-seed set $\mathcal{S}_{\text{elig}}$ and 40-scenario panel $\mathcal{E}_{\text{elig}}$, selecting

$$\mathcal{T} = \left\{ i : \hat{p}_{i,\text{elig}}^0 = \frac{1}{|\mathcal{S}_{\text{elig}}||\mathcal{E}_{\text{elig}}|} \sum_{s \in \mathcal{S}_{\text{elig}}} \sum_{e \in \mathcal{E}_{\text{elig}}} z_{i,s,e}^0 \geq 0.20 \right\}. \quad (2)$$

The inclusive threshold requires at least 24 successes among 120 rollouts and reduces severe floor censoring without using donor outcomes. It selects 14 tasks and registers $14 \times 13 = 182$ ordered non-self pairs. The 26 excluded candidates remain in the provenance release; Appendix A.1 gives the manifests, roster, and suite composition. Table 1 reports receipt-validated sample rather than planned counts.

Let $\mathcal{D}_i$ and $\mathcal{D}_j$ be the registered pools of 20 demonstration episodes for target i and donor j . Every ordered pair uses Diffusion Policy [2], trained from initialization for 10,000 updates with batch size 32. The target-only condition draws only from $\mathcal{D}_i$; mixed training keeps all episodes available

Table 1: Receipt-validated scale at each dataset-construction stage.

Stage	Task or execution scale	Audited records
Source registry	40 candidate tasks	800 episodes; 133,328 steps
Benchmark roster	14 selected tasks; 182 ordered pairs	280 episodes; 33,746 steps
Checkpoint calibration	240 checkpoint evaluations	2,400 rollouts
Eligibility screen	120 target-only runs	4,800 rollouts
Primary matrix	546 mixed + 42 target-only runs	5,880 rollouts; 182/182 cells complete

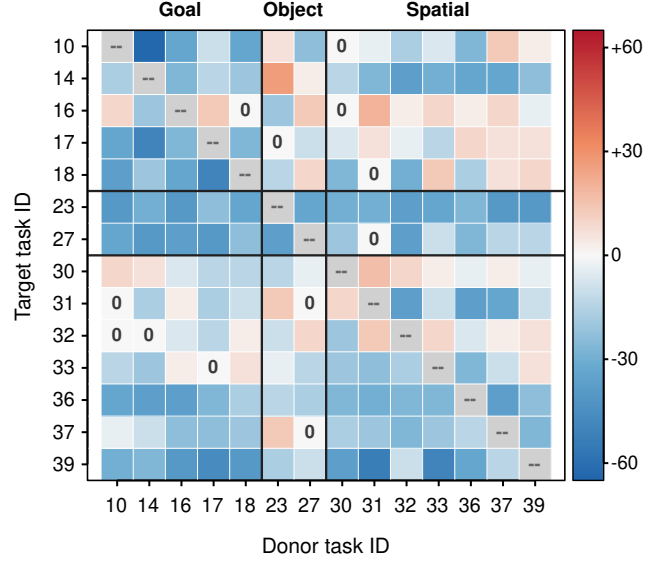
but replaces exactly 10% of target window draws with donor window draws. All other training choices and the total number of updates are fixed. The contrast therefore measures a fixed-budget replacement intervention: donor exposure jointly with reduced target exposure, not additive data or fewer unique target episodes. Exact sampling and random-stream rules appear in Appendix A.2.

The primary matrix shares its training seeds with the eligibility audit: $\mathcal{S}_{\text{mat}} = \mathcal{S}_{\text{elig}}$. Utility uses the frozen t_i^* and ten fixed scenarios $\mathcal{E}_{\text{util}} \subset \mathcal{E}_{\text{elig}}$ for every mixed-versus-target-only comparison. All three scenario sets and their relationships are fixed before donor outcomes are observed. Thus no mixed outcome affects roster construction, but the released matrix is selection-conditioned on target-only performance rather than an unconditioned estimate over the 40-task candidate pool. Let $z_{i,j,s,t,e}^{\text{mix}}$ denote binary mixed-policy success. After freezing t_i^* , abbreviate $z_{i,j,s,e}^{\text{mix}} := z_{i,j,s,t_i^*,e}^{\text{mix}}$, and define the matched scenario-level contrast $d_{i,j,s,e} := z_{i,j,s,e}^{\text{mix}} - z_{i,s,e}^0$. A complete primary-matrix cell has stored utility

$$U(i,j) = \frac{1}{|\mathcal{S}_{\text{mat}}||\mathcal{E}_{\text{util}}|} \sum_{s \in \mathcal{S}_{\text{mat}}} \sum_{e \in \mathcal{E}_{\text{util}}} d_{i,j,s,e}. \quad (3)$$

Equation 3 pairs the contrast by target, seed, checkpoint, and scenario; positive and negative values mean higher and lower completion than target-only training. The release and scorer store this rate difference in $[-1, 1]$; reported utilities, unnormalized errors, and regrets are multiplied by 100 and expressed in pp, whereas normalized and ranking metrics remain dimensionless. Each complete cell therefore changes in increments of $100/30 = 3.33$ pp. Valid task failures have success zero, invalid or absent rollouts remain missing, and $U(i,i) = 0$ is only a storage convention. We denote the target-only rate on this same seed-scenario panel by $p_{i,\text{util}}^0$; it is distinct from the eligibility-panel rate used to select the roster.

Each cell links manifests, configuration and checkpoint hashes, aggregate window-draw counts, and rollout outcomes. The public tables contain 588 complete canonical runs and their 5,880 observed rollouts; invalid or absent executions do not enter a released utility. The v1.0.0 bundle audit checks production schemas, record counts and keys, run-rollout joins,

**Figure 2: Directed target-donor utility matrix in percentage points.**

the ten-scenario panel, file hashes, and recomputes every utility. Execution used the frozen `libero.discovery_14x14.v1` protocol and schema version 1; exact schemas, checksums, and integrity rules appear in Appendix A.7 and the release manifest.

Figure 2 visualizes directed utility with targets in rows and donors in columns. Each off-diagonal cell is the mixed-minus-target-only completion-rate difference averaged over three seeds and ten matched scenarios: red means improvement, blue degradation, and 0 an exact tie. Gray “_” cells mark structural self-pairs, and heavy rules separate Goal, Object, and Spatial suites. Predominantly blue cells, punctuated by sparse red ones, show generally harmful yet target-dependent and directional transfer.

3.2 Dataset Characterization and Validation

The target-only screen selects a 14-task roster with 280 episodes and 33,746 steps from the 40-task candidate pool. It contains five LIBERO-Goal, two LIBERO-Object, and seven LIBERO-Spatial tasks, but no LIBERO-10 task. This is a material coverage bias: the matrix and benchmark describe the selected roster, not the full candidate pool or LIBERO broadly; the 26 excluded candidates remain available for provenance analysis. Table 2 reports the audited checkpoint distribution. The 5,000- and 7,000-update checkpoints account for 32 of 40 candidates and 11 of 14 eligible tasks. At the selected checkpoints, calibration contains 105 successes among 400 rollouts; 19 targets have success rates in $[0.3, 0.7]$, whereas 21 remain below 0.3 at every candidate checkpoint.

Table 2: Distribution of selected checkpoints for candidate and eligible tasks.

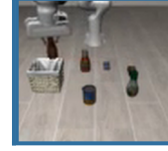
Checkpoint (updates)	5k	6k	7k	8k	9k	10k
Candidate tasks	24	2	8	4	1	1
Eligible tasks	6	1	5	2	0	0

Checkpoint selection is coarse: 26 tasks have tied candidates, including 16 with identical counts at all six checkpoints. Leave-one-scenario-out retains the checkpoint in 347 of 400 trials, and paired bootstraps reselect it with mean frequency 0.7317. These diagnostics measure calibration sensitivity, not utility-scenario performance, cross-seed reliability, or final convergence. Appendix A.1 gives the task-level results and synchronized perturbation procedure.

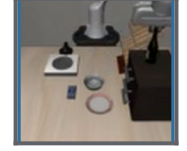
We characterize the audited matrix through signed utility, target-row range, donor-task column heterogeneity, suite blocks, and reciprocal asymmetry $U(i, j) - U(j, i)$, using frozen registry order. Zero, missing, and diagonal cells remain distinct. Because target i 's feasible reported range is $[-100p_{i, \text{util}}^0, 100(1 - p_{i, \text{util}}^0)]$ pp, its utility-panel target-only headroom accompanies every row summary. Practical equivalence and sign assessment remain separate: the 5-pp decision tolerance ($\epsilon = 0.05$ in scorer units) treats near-best actions as interchangeable; it does not establish a zero effect. Paired resampling uses common scenario indices across all donor contrasts in a target row, so the shared target-only outcome is resampled once rather than independently for each cell. This preserves row dependence and limits the resulting assessment to the observed seed-scenario panel.

Across the 182 cells, mixed policies complete 1,730/5,460 rollouts (31.7%) versus 2,548/5,460 (46.7%) for the pair-weighted target-only reference, giving mean utility of -15.0 pp. Forty cells are positive, 12 are zero, and 130 are negative; 30 lie within ± 5 pp. Target-only rates span 20.0%–73.3%, and within-row donor spreads span 16.7–76.7 pp. Across 91 reciprocal pairs, mean absolute asymmetry is 22.3 pp and reaches 56.7 pp. Thus aggregate harm coexists with substantial target dependence and directionality. These are selection-conditioned finite-panel summaries, not population-level effects. Figure 2 shows the complete directed pattern, while Figure 3 connects its largest positive and negative cells to the corresponding tasks.

With $|S_{\text{mat}}| = 3$, repeatability diagnostics describe the finite construction panel rather than population-wide training variation. Across the 42 target-by-seed-pair comparisons, the target-macro mean Spearman correlation between 13-donor rankings is 0.083 (median 0.085; range -0.458 to 0.485). Top-five sets share 1.95 of five donors on average, using frozen registry order to break ties. Leaving out each seed reproduces the full-panel selected action in 21/42 cases (50.0%) and selects an action within 5 pp of the full-panel optimum in 27/42 cases (64.3%). These results show limited construction-seed stability. A separate held-out-seed validation completed after matrix freezing and confirmed the primary matrix's qualitative conclusions.

BENEFICIAL PAIR**+26.7 pp****DONOR 23****Pick up the BBQ sauce and place it in the basket****TARGET 14****Put the wine bottle on top of the cabinet**

9:1 mix

**Target-only 56.7% → Mixed 83.3%****HARMFUL PAIR****-63.3 pp****DONOR 14****Put the wine bottle on top of the cabinet****TARGET 10****Put the bowl on the plate**

9:1 mix

**Target-only 70.0% → Mixed 6.7%****Figure 3: Actual task pairs with positive and negative donor utility.**

Semantic and artifact validity are assessed separately. The frozen LIBERO goal predicate supplies the binary label; an independent, suite- and outcome-stratified review checks videos and traces under a fixed completion codebook. Reviewers are blinded to pair utility and the automatic label, receive repeated and edge cases, and adjudicate conflicts under the same codebook; reviewer-level records are not part of v1.0.0. The public artifact audit instead checks schemas, registries, run-rollout joins, scenario coverage, file hashes, path hygiene, and utility recomputation. It accepts all 588 released training records and 5,880 rollouts in Table 1, with all 182 cells complete and recomputed at denominator 30; valid unsuccessful executions count as failures rather than missing data. Sampling declarations and file-level results appear in Appendices A.2 and A.7.

The resulting utility values are conditioned on the selected roster, learner, mixing ratio, training compute, checkpoint, evaluation scenarios, and simulator. The roster is itself conditioned on target-only eligibility, and all 13 donor contrasts within a target-seed row share one target-only policy and are thus statistically dependent. Accordingly, the matrix supports protocol-specific donor decisions and should not be interpreted as learner-independent task similarity.

4 Benchmark and Evaluation

This section evaluates how the directed utility reference supports decisions before the outcome of a new target–donor intervention is known. We separate the benchmark contract from the evaluated implementations so that performance differences cannot be attributed to different information access or scoring rules. A shared protocol fixes permitted inputs, actions, and metrics; role-disjoint splits isolate unseen task identities; and the resulting evidence compares safe donor choice with utility recovery under limited measurements.

4.1 Benchmark Protocol

Both interfaces model the point at which a practitioner has target and donor descriptions but has not observed mixed-training outcomes for the held-out target. Every utility involving that identity is excluded from model fitting and remains available only to the scorer. The interfaces differ in whether the method must act without any target-row utility or may acquire a small number of cells before completing the row. Table 3 summarizes their permitted inputs, actions, and scores.

Donor Selection for New Targets selects a donor or NoDonor for a held-out identity. **Budgeted Utility Completion** evaluates recovery and decisions after limited measurements. *Data-only* provides language, trajectories, statistics, and metadata; *Calibration-aware* adds target-only and checkpoint features. Neither reveals held-out mixed outcomes.

For donor selection on held-out target i , the method submits scores for all 13 donors after every utility involving i is removed from fitting. The scored action set is $\mathcal{A}_i^{\text{valid}} = \{\text{NoDonor}\} \cup \{j \neq i : U(i, j) \text{ is valid}\}$; the diagonal is never submitted or scored, and NoDonor has predicted and observed utility zero. The scorer uses the submitted utility score for selection, magnitude error, and ranking; NoDonor wins a tie at zero, while donor–donor ties follow frozen registry order. Decision Utility is the measured utility of the selected action. With measured row optimum U_i^* , practical regret is $R_i^\epsilon = \max(0, U_i^* - U(i, \hat{a}_i) - \epsilon)$ for $\epsilon = 0.05$ in scorer units, equivalent to 5 pp. We report Decision Utility, regret, and target-macro MAE in pp, and the fraction improved as a percentage; Spearman and NDCG@5 rank NoDonor with all valid non-self donors and remain dimensionless.

For budgeted completion, each trial partitions a target’s 13 donor cells into a five-cell query pool $\mathcal{Q}_r$ and an immutable eight-cell scoring mask Ω_r . We enumerate all $\binom{13}{5} = 1,287$ pools per target (18,018 trials) and use nested budgets $q \in \{0, 1, 2, 3, 5\}$. A strategy may reveal cells only from $\mathcal{Q}_r$, never from Ω_r [6]. At budget q , normalized Frobenius error is

$$\text{NFE}_{r,q} = \frac{\|P_{\Omega_r}(\hat{U} - U)\|_F}{\|P_{\Omega_r}(U)\|_F}. \quad (4)$$

Queried cells never enter this prediction-error denominator; NFE is undefined when its reference norm is zero, although no released trial has that edge case. For the decision metric, queried cells use measured utility and unqueried cells use submitted predictions. Both NFE and ϵ -regret are reported

by budget and as trapezoidal curve AUC divided by the five-cell budget span. Regret and regret AUC use pp, whereas NFE and NFE AUC remain dimensionless. Appendix A.3 specifies the remaining edge cases, costs, and audit records.

4.2 Experimental Setup and Baselines

The evaluation targets task-level generalization rather than interpolation among pairs whose endpoint identities were already observed during fitting. Accordingly, fourteen outer folds exclude each held-out identity in both target and donor roles; methods fit the 156 remaining pairs, with hyperparameters selected by an inner role-disjoint loop. This follows task-level evaluation in Meta-World and OGB [7, 21]. Results macro-average targets and use 10,000 complete-target bootstrap resamples; Appendices A.6 and A.3 give full details.

Decision references are NoDonor, *Random donor*, and *Historical donor average*. Outcome-free baselines use instruction cosine or directed trajectory coverage [15, 18]. A role-aware ridge predictor combines language, DINOv2 visual, state-action, and task features [14]; its calibration-aware variant adds target-only features.

For completion, random and diversity-first query rules share a rank-3 biased low-rank predictor and differ only in query order. DataMIL, CUPID, and gradient task-affinity require signals outside the frozen boundary and are not directly comparable [1, 3, 10]. Appendix A.4 gives implementation and available compute details; v1.0.0 lacks aggregate accelerator-hours.

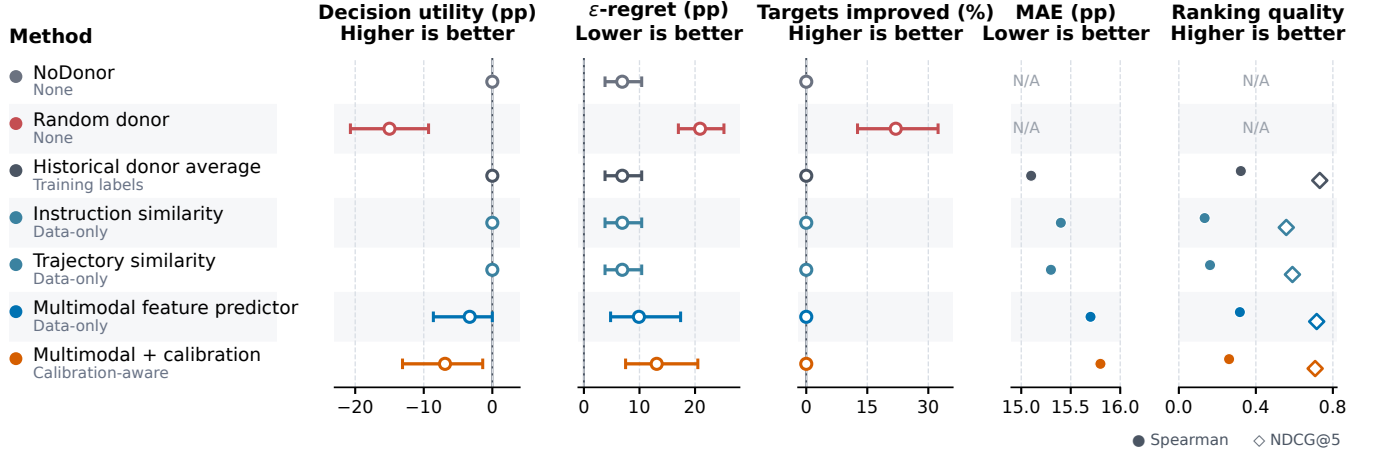
4.3 Results

We organize the results by downstream decision relevance. Donor Selection for New Targets tests whether outcome-free features can improve on the safety of target-only training for unseen identities. Budgeted Utility Completion tests how limited measurements change both utility recovery and donor choice. Decision Utility and ϵ -regret are primary, while prediction and ranking metrics explain where useful signal is or is not converted into an action. Confidence intervals resample complete held-out targets, preserving dependence among cells in the same target row.

Donor Selection for New Targets. NoDonor is the strongest tested safety baseline despite 40 positive cells. Figure 4 reports the role-disjoint experimental results for the evaluated baselines, contrasting decision performance with prediction and ranking diagnostics. Historical donor average, instruction similarity, and trajectory similarity all choose NoDonor, obtaining Decision Utility 0.0 pp and ϵ -regret 6.9 pp; this is safer than random donor selection, whose expected Decision Utility is -15.0 pp. The multimodal predictor selects a donor for two targets; both selections are harmful, yielding -3.3 pp $[-8.6, 0.0]$. Adding calibration leads to six donor selections, five harmful and one neutral, and lowers Decision Utility to -6.9 pp $[-13.1, -1.4]$. Historical and multimodal predictors show the strongest ranking signal even though their final actions are not beneficial. This gap between ranking quality and Decision Utility motivates

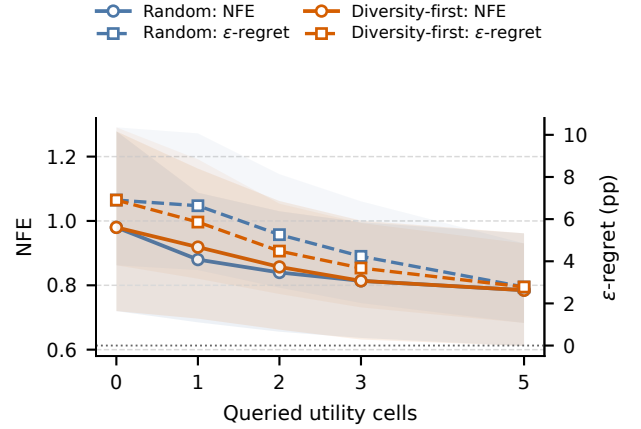
Table 3: Inputs, actions, and metrics for the two benchmark tasks.

Task	What the method sees	What the method does	What the benchmark scores
Donor Selection for New Targets	Data-only or Calibration-aware features for the held-out target and 13 candidate donors; no utility involving the target	Predict 13 utilities; select one donor or NoDonor	Hidden 13-cell target row: Decision Utility, ϵ -regret, and fraction improved
Budgeted Utility Completion	The same feature view plus q measured cells from a five-cell query pool, $q \in \{0, 1, 2, 3, 5\}$	Predict the eight cells in Ω_r ; choose a donor using measured values and remaining predictions	Fixed hidden mask Ω_r : NFE and ϵ -regret at each budget and as curve AUC

**Figure 4: Role-disjoint experimental results for the evaluated donor-selection baselines across decision and diagnostic metrics.**

objectives aligned with the downstream action and better calibration of the NoDonor threshold. For target 14, both learned predictors choose donor 33 (−30.0 pp) instead of donor 23 (+26.7 pp); for target 16, all five selectors abstain and miss donor 31 (+20.0 pp). These descriptive cases demonstrate measured positive transfer that current selectors miss; full actions appear in Appendix A.5. The gap motivates target-directed representations, decision-aware objectives, and uncertainty-aware abstention. TUTOR-BENCH provides a common outcome reference for comparing these designs under one role-disjoint protocol.

Budgeted Utility Completion. Diversity-first querying improves decisions before full-row recovery; Figure 5 reports the two completion baselines’ experimental results, contrasting NFE with downstream ϵ -regret. Revealing all five query cells reduces NFE from 0.980 to 0.785 and ϵ -regret from 6.9 to 2.8 pp; Decision Utility rises from 0.0 to 5.8 pp. At $q = 2$, diversity-first regret is 4.5 pp versus 5.3 pp for random querying, a paired difference of −0.779 pp (95% CI [−1.593, −0.032]); Decision Utility is higher by 0.965 pp ([0.005, 1.991]). Its regret AUC is also lower (4.419 versus 4.897 pp; difference −0.478 pp, [−0.962, −0.042]), whereas the dimensionless NFE AUC difference is unresolved (0.01123, [−0.00678, 0.03295]). The measured advantage is therefore

**Figure 5: Budgeted-completion experimental results for the random-query and diversity-first baselines.**

decision-specific; full-row recovery remains unresolved. For selector design, this favors coupling utility prediction with a decision-aware query policy rather than treating reconstruction error as the sole acquisition objective. The counterfactual

charge is nine runs and 90 rollouts at $q = 2$, versus 18 runs and 180 rollouts at $q = 5$; these counts price reuse of measured cells. Calibration also fails to rescue selection: Decision Utility falls from -3.3 to -6.9 pp. The results bound the tested outcome-free features under a role-disjoint split.

5 Access and Limitations

TUTOR-BENCH v1.0.0 is publicly available through GitHub and Hugging Face, pinned to the immutable revisions reported in Appendix A.7. GitHub provides code, schemas, documentation, the CPU-only scorer, and compact benchmark artifacts; Hugging Face carries the full records, features, and predictions. Code, schemas, and documentation use Apache 2.0, while project-generated data and benchmark artifacts on either host use CC BY 4.0. The release includes per-file checksums and Dataset and Benchmark Cards. It excludes raw LIBERO demonstrations, simulator assets, videos, action traces, and model checkpoints; stable identifiers, hashes, attribution, and reconstruction instructions preserve the upstream boundary. Appendix A.7 gives the exact artifact layout and audit scope.

Each host-level `MANIFEST.json` records relative paths, SHA-256 hashes, byte sizes, and record counts where applicable, with `SHA256SUMS` as the file inventory. The completed public-bundle audit verifies manifest coverage, path hygiene, production-schema conformance, primary-key uniqueness, roster selection, run-rollout joins, ten-scenario coverage, all 182 recomputed utilities, the matrix hash, completion totals, zero query-scoring overlap, and equality of files shared across the two hosts. A compact synthetic fixture and CPU scorer tests support lightweight validation. Full training reproduction is separate because upstream demonstrations, the simulator, and checkpoints are not redistributed.

All conclusions are conditioned on a target-only screen that selects 14 of 40 LIBERO tasks, one learner, a 9:1 exposure ratio, target-specific checkpoints, three training seeds, ten utility scenarios, and one simulator. No LIBERO-10 task enters the selected roster, so the results do not describe the 26 excluded tasks or LIBERO broadly. The utility scenarios are a subset of those used for eligibility, making target-only rates selection-conditioned; ten scenarios also give coarse seed-specific estimates, and donors in one row share a target-only reference. The primary evaluation has only 14 target-level test units and a known donor library, so it does not evaluate unseen donors or establish transfer across learners, ratios, embodiments, or real scenes. Mixed and target-only training match total updates rather than target exposure: both retain all 20 target episodes, while mixed training replaces 10% of target draws. The benchmark can price avoided task-pair interventions, but cannot show that fewer target demonstrations suffice.

6 Ethics Statement

This study adds simulator-derived outcome annotations to LIBERO, recruits no external participants, and collects no

new personal or demographic information; the authors' manual audit evaluates simulated task completion rather than people. Hosting-platform access, terms of service, and non-commercial research purpose are not treated as redistribution permission. Version 1.0.0 therefore excludes upstream demonstration bytes and simulator assets, retaining stable identifiers, checksums, attribution, and reconstruction instructions. The target-only screen selects 14 of 40 LIBERO tasks and excludes every LIBERO-10 task, so the matrix reflects selection, simulator, learner, and label biases and supports no claim about human demographic fairness. Utility is a protocol-conditioned contrast, not an intrinsic dataset ranking or evidence of physical-robot safety. The release preserves negative and missing outcomes and reports per-file sizes, rollout wall times, the feature-extraction device, and the completion job's runtime; it does not claim a complete training/rollout accelerator-hour or energy account. Appendix A.8 gives the exact licensing, governance, and resource scope.

7 Conclusion

TUTOR-BENCH turns a data-allocation question exposed by VLA scaling into a directed, measurable decision. Starting from four LIBERO suites, a target-only screen selects a final 14-task roster from 40 candidates before any donor outcome is observed. Across the 182 ordered pairs within this roster, each donor-task intervention compares 9:1 mixed training with scenario-matched target-only training at a checkpoint fixed before donor outcomes are observed. The resulting records support 14-fold role-disjoint Donor Selection for New Targets and Budgeted Utility Completion with integer query budgets. In this selected finite panel, mixed policies complete 31.7% of rollouts versus 46.7% for target-only policies; 130 of 182 measured cells are negative, whereas 40 are positive. Utilities range from -63.3 to $+26.7$ pp, and the mean absolute reciprocal difference is 22.3 pp, showing that donor effects are strongly target-dependent, but the 0.083 mean seed-pair rank correlation also shows that their ordering is not yet stable across training seeds. In the budgeted-completion task, diversity-first querying lowers regret AUC from 4.897 to 4.419 pp relative to random querying, while the dimensionless NFE AUC difference is unresolved. The result supports active measurement for donor decisions, not a general claim of better matrix reconstruction.

The benchmark measures reuse of training exposure under one declared learner and simulator protocol. It does not assign intrinsic value to donor tasks or show that fewer unique target demonstrations can be collected. Extending the outcome reference to jointly language-conditioned VLA learners, other ratios and embodiments, and explicit target-pool budgets is therefore an important next step.

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A Registry, Protocol, and Artifact Details

This appendix fixes the implementation and scoring details needed to reproduce the benchmark. It does not define a different utility estimand, and it is not the sole support for any headline empirical claim: dataset construction and quality are reported in Section 3, and benchmark comparisons remain in Section 4.3.

A.1 Registry and Checkpoint-Selection Audit

Benchmark construction starts from LIBERO: the candidate pool contains 40 tasks drawn from four suites with ten tasks each. A deterministic manifest selects 20 demonstration episodes per candidate task, for 800 globally unique episodes and 133,328 recorded steps. Each task record contains the immutable task ID, suite, language instruction, environment horizon, and source environment identifier. Each episode record contains its source identifier, task foreign key, length, and selected-set membership; source-file checksums pin the upstream bytes. Target-only screening selects the final 14-task benchmark roster, which contains 280 episodes and 33,746 steps. Candidate order is inherited by the roster and used for tables, heatmaps, and deterministic tie-breaking; no mixed outcome may alter membership or order.

We use three fixed scenario sets: $\mathcal{E}_{\text{cal}}$ for ten checkpoint-calibration scenarios, $\mathcal{E}_{\text{elig}}$ for the 40-scenario roster-eligibility panel, and $\mathcal{E}_{\text{util}}$ for ten fixed utility-estimation scenarios. Thus $\mathcal{E}_{\text{cal}} \cap \mathcal{E}_{\text{elig}} = \emptyset$ and $\mathcal{E}_{\text{util}} \subset \mathcal{E}_{\text{elig}}$. Calibration uses one fixed training seed s_{cal} . Eligibility and primary-matrix construction share the three-seed set $\mathcal{S}_{\text{elig}} = \mathcal{S}_{\text{mat}}$.

Section 3 defines target-only success, the checkpoint estimator in Eq. (1), and the inclusive roster rule in Eq. (2). Section 3.2 reports the audited checkpoint distribution and calibration-resolution diagnostics; these diagnose the finite calibration panel rather than utility-evaluation success, cross-seed reliability, or final-convergence performance.

The eligibility audit contains 969 successes among 4,800 target-only rollouts. Applied after calibration but before mixed training, Eq. (2) selects the final 14-task roster, containing 794 successes among 1,680 eligibility rollouts. The selected roster comprises five LIBERO-Goal, two LIBERO-Object, and seven LIBERO-Spatial tasks; no LIBERO-10 task reaches the threshold. Table 4 gives the task definitions; task 32 is included because its eligibility rate equals 0.20. No mixed-policy checkpoint, metric, or rollout is observed before roster selection.

Checkpoint sensitivity is computed with synchronized perturbations. The leave-one-scenario-out audit deletes the same calibration scenario from all six checkpoint candidates for a target; it preserves the original selection in 347 of 400 target-deletion trials, with 25 targets unchanged under every deletion. The paired scenario bootstrap applies one resampled index vector to all six candidates for each of 10,000 replicates per target. Its mean reselection frequency is 0.7317, and the modal checkpoint agrees with the original for 37 of 40 targets. Twenty-six targets have tied candidates and

16 have identical success counts at all six checkpoints, so the frozen earlier-checkpoint tie rule contributes to the observed stability. $\mathcal{E}_{\text{cal}}$ is disjoint from $\mathcal{E}_{\text{elig}}$, while $\mathcal{E}_{\text{util}} \subset \mathcal{E}_{\text{elig}}$ is used for both target-only and mixed policies. Version 1.0.0 records the calibration and eligibility summaries and the utility-scenario identifiers; the top-level manifests checksum the corresponding public tables. Thus utility evaluation is disjoint from checkpoint calibration but not from the target-only outcomes used to select the roster; the released estimand is selection-conditioned.

A.2 Sampler, Pairing, and Retry Semantics

The exact 9:1 exposure specification uses a ten-update cycle. Eight batches contain 29 target and 3 donor window draws, and two contain 28 target and 4 donor window draws. A complete cycle therefore contains 288 target and 32 donor window draws; over 10,000 updates this yields 288,000 target and 32,000 donor window draws. The target-only policy uses 32 target window draws in every batch, for 320,000 target window draws. The released `data/training_runs.jsonl` records the condition, task pools, checkpoint, and aggregate draw counts for all 588 complete runs. Version 1.0.0 validates these records against the production schema, but does not publish per-update sampler traces or per-episode realized draw counts. Within each target or donor pool, action windows are sampled uniformly with replacement, so an episode’s marginal draw probability is proportional to its number of windows. Independent PyTorch generators use keys $s + 1009$ and $s + 2003$ for target and donor windows. Batches emit target then donor draws without reshuffling; runs use `num_workers=0`, target-only normalizers, initialization from scratch, and no resume. These mechanics define the private execution protocol; the public bundle exposes the task pools and aggregate 288,000/32,000 mixed-run totals rather than the ordered sampler trace.

Mixed and target-only rollout records are paired only when target, training seed, target-specific checkpoint, and utility-scenario identifier match. For every valid pair, the scenario-level contrast is $d_{i,j,s,e} := z_{i,j,s,e}^{\text{mix}} - z_{i,s,e}^0$. The same join recomputes every $d_{i,j,s,e}$ and primary utility $U(i,j)$ over $\mathcal{S}_{\text{mat}}$ and $\mathcal{E}_{\text{util}}$. The shared target-only record is referenced rather than duplicated across the 13 donor-task contrasts in one target-seed row. A common seed identifier supplies a blocking key. Version 1.0.0 includes no RNG manifest or execution trace sufficient to establish stronger coupling of initialization, augmentation, action sampling, or simulator randomness, so none is inferred from that identifier.

A valid policy execution that fails the frozen simulator goal predicate is a task failure and receives success 0. An absent receipt, corrupt trace, infrastructure exception, or invalid simulator execution is missing. Retries retain all attempt records, but exactly one attempt may become canonical for a task pair, and a failed attempt is never silently overwritten. Utilities may be computed only from valid matched rollout

Table 4: Frozen 14-task benchmark roster and released language instructions.

ID	Task (language instruction)
10	put the bowl on the plate
14	put the wine bottle on top of the cabinet
16	turn on the stove
17	put the bowl on the stove
18	put the bowl on top of the cabinet
23	pick up the bbq sauce and place it in the basket
27	pick up the butter and place it in the basket
30	pick up the black bowl next to the cookie box and place it on the plate
31	pick up the black bowl in the top drawer of the wooden cabinet and place it on the plate
32	pick up the black bowl on the ramekin and place it on the plate
33	pick up the black bowl on the stove and place it on the plate
36	pick up the black bowl next to the plate and place it on the plate
37	pick up the black bowl next to the ramekin and place it on the plate
39	pick up the black bowl on the wooden cabinet and place it on the plate

pairs under a cell-validity rule frozen before aggregation; missing outcomes are neither converted to failures nor imputed as zero utility. The public v1.0.0 tables contain only canonical complete records and their hashes, not attempt-level retry histories. A complete primary-matrix cell has denominator 30 under the frozen protocol.

A.3 Scorer Conventions

One prediction record is keyed by benchmark version, split or fold, input view, target-task ID, donor-task ID, method version, and, for acquisition, trial and budget. Section 4.1 defines the shared-score rule, valid action set, diagonal exclusion, tie handling, and 5-pp practical regret. The corresponding practical-equivalence set $\mathcal{A}_i^c$ contains every action within 5 pp of the measured row optimum.

MAE and RMSE are computed on valid donor-task cells, macro-averaged over valid target rows, multiplied by 100, and reported in pp. Rank metrics use NoDonor and all valid non-self donor tasks in the declared action set. Spearman correlation uses midranks, and Kendall correlation uses the tie-adjusted τ_b . For NDCG@5 (normalized discounted cumulative gain at rank 5), row-wise relevance is

$$q_i(a) = U(i, a) - \min_{b \in \mathcal{A}_i^{\text{valid}}} U(i, b), \quad (5)$$

and DCG uses gain $2^{q_i(a)} - 1$ with logarithmic rank discount over $\mathcal{A}_i^{\text{valid}}$. A row with zero ideal DCG assigns NDCG 1 because every action is tied. Mean reciprocal rank (MRR) is the reciprocal rank of the first action in the practical-equivalence set $\mathcal{A}_i^c$, using the 5-pp tolerance. Metric denominators and the number of excluded rows are reported with every aggregate. Equation 5 fixes the relevance shift used by NDCG@5.

Table 5 reports the two pair-prediction controls omitted from the main decision table. Both assign NoDonor a positive score and therefore select NoDonor for all 14 targets, giving Decision Utility 0.0 pp and ϵ -regret 6.9 pp. Their purpose is only to show how much feature-based methods improve utility magnitude and ranking over constant predictions.

Table 5: Pair-prediction results for constant diagnostic controls.

Control	MAE (pp)	RMSE (pp)	Spearman	NDCG@5
Predict zero utility	18.9	21.7	N/A	0.574
Overall training mean	15.2	17.8	0.223	0.574

For acquisition, each 13-cell target row is partitioned into a five-cell query pool and an eight-cell scoring mask. All $\binom{13}{5} = 1,287$ query pools are enumerated per target. Query histories are nested across $q \in \{0, 1, 2, 3, 5\}$, and metrics at every budget use the same Ω_r . Equation (4) and the surrounding protocol in Section 4.1 define NFE, query-cell exclusion, revealed versus predicted decision values, and budget-curve AUC. A zero reference norm would retain the trial for RMSE and decision metrics while marking NFE undefined; none of the 18,018 released trials has this edge case. Each queried cell costs three mixed runs and 30 evaluation rollouts. The first query also charges three shared target-only runs and 30 rollouts; later queries reuse that reference. Thus $q = 2$ costs nine runs and 90 rollouts, whereas $q = 5$ costs 18 runs and 180 rollouts. The completion scorer uses protocol ID `budgeted-utility-completion-14-v1`. All 18,018 trials contain five query cells and eight scoring cells. Intervals use 10,000 complete-target bootstrap resamples with seed 20270727 after averaging masks within each target. Version 1.0.0 lists the scorer tests, completion audit, masks, trial metrics, and summaries in its top-level manifest and SHA256SUMS; no shortened hash is authoritative.

A.4 Baseline Implementation Details

Instruction similarity uses cosine similarity between 384-dimensional ℓ_2 -normalized all-MiniLM-L6-v2 embeddings, frozen at revision 1110a243fdf4; the release records the full revision. Trajectory similarity robustly scales state, action, summary, gripper, and temporal blocks within each fitting fold, then measures directed coverage by the mean distance

from each target episode to its nearest donor episode. A nonnegative-slope ridge calibrator maps each similarity to utility units using outer-training pairs only.

The Multimodal feature predictor combines language, DINOv2 visual, state-action, episode-length, horizon, and suite descriptors. PCA and scalar normalization are fitted within each fold. For task vector x_i , the ordered pair representation is $[x_i, x_j, |x_i - x_j|, x_i \odot x_j]$. The Calibration-aware variant adds the six-checkpoint target-only curve and selected checkpoint from seed 2027 on scenarios 0–9; endpoint scenarios and mixed outcomes are never inputs. Completion fits a rank-3 biased low-rank predictor with unit regularization on the 156 outer-training pairs. Random query uses a SHA-256-induced ordering with acquisition seed 2027. Diversity-first query begins with the donor factor farthest from the five-cell pool centroid and then applies maximin traversal. Both rules update the same held-out row intercept and latent factor from only the revealed utilities.

Similarity penalties $\{0, 0.01, 0.1, 1, 10\}$ and ridge penalties $\{0.01, 0.1, 1, 10, 100\}$ are selected by nested role-disjoint validation, ordered by ϵ -regret, Decision Utility, MAE, and then the larger penalty on an exact tie. The deterministic baselines have no model seed. Intervals use 10,000 complete-target bootstrap resamples with seed 20270727. The feature cache contains 280 demonstrations and 33,746 state-action steps; visual features were extracted on one RTX 3090. Release records retain the split, feature-inventory, source-file, prediction, and score hashes.

A.5 Positive Transfer Case Studies

Table 6 reports the two largest measured positive cells discussed in Section 4.3. The cases are selected descriptively and are not used for inference; each method entry gives its selected action and measured utility.

A.6 Splits and Leakage Controls

The primary split defines 14 outer folds in frozen registry order. Fold i holds out task i , removes every utility with target i or donor i , and fits on all $13 \times 12 = 156$ directed non-self pairs among the remaining tasks. It scores the 13 outgoing cells of i against NoDonor. Every target row and every directed pair is therefore scored exactly once, and the held-out task supplies no utility label in either role.

Hyperparameter selection uses a nested leave-one-task-out loop inside each outer fold. An inner fold removes its validation identity in both roles, scores that identity against the remaining tasks, fits on $12 \times 11 = 132$ pairs, and contributes one target-level validation result. Hyperparameters minimize the target-macro criterion across all 13 inner folds; the selected configuration is then refit on the 156 outer-training pairs.

Data-only features are derived from source demonstration episodes and task metadata; Calibration-aware features additionally use target-only calibration. Feature normalization, label-dependent transforms, and hyperparameters are fitted on outer-training identities and selected in the inner loop.

Mixed-policy artifacts, utilities incident to the held-out identity, and scoring-mask cells remain unavailable to a method. In acquisition, only task pairs listed in the cumulative query history may be revealed, and the immutable scoring mask is never exposed. The released `splits.json` fixes outer and inner folds, while compressed completion masks and trial metrics record query pools, ordered query histories, scoring cells, and hashes. The completion audit reports zero query-scoring overlap and zero held-out-identity training cells; v1.0.0 does not include online scorer access logs or a sealed-evaluation service.

A.7 Schema and Reproducibility Contract

Table 7 gives the physical v1.0.0 artifact layout. The five core JSONL tables are governed by the production schemas in `schema/catalog.json`; benchmark arrays and outputs are bound by their manifests and audit records. Machine-local paths are not valid identifiers, and the released null semantics never equate unavailable data with failure or zero utility.

Section 5 reports the manifest fields, completed public-bundle audit, lightweight validation fixture, and boundary on full training reproduction. Table 7 provides the file-level mapping needed to interpret those guarantees.

The paper’s author-conducted semantic review is separate from this artifact audit. Version 1.0.0 contains no stratified reviewer manifest, per-rater decisions, adjudication records, or agreement-recomputation bundle; the public audit therefore certifies artifact integrity, not semantic agreement with the simulator success predicate.

The code bundle contains the Dataset and Benchmark Cards, data dictionary, licenses and component inventory, known issues, reconstruction guide, scorer, tests, and compact benchmark summaries. Hugging Face carries the full compressed rollout, feature, prediction, and completion artifacts. Neither top-level manifest records a container digest, code commit, host revision, dependency lock, aggregate accelerator-hours, or energy estimate. The deposited GitHub tree is fixed at 678c9c98261b; its annotated v1.0.0 tag resolves to that commit. The Hugging Face tree is fixed at 90db1432d69a; its annotated v1.0.0 tag resolves to that content commit. Host revisions are deposition metadata, not manifest fields.

A.8 Ethics, Licensing, and Governance Details

The author-conducted semantic review is limited to simulator rollout records. Reviewers are not asked to infer intent, identity, demographic attributes, or sensitive characteristics, and no external annotator labor or compensation is involved. As noted above, reviewer-level materials are not distributed in v1.0.0 and are outside the public artifact audit’s claim.

The component license inventory records the copyright holder, authoritative evidence, license or applicable terms, required attribution and notices, modification status, and redistribution decision for each artifact class. Technical access through GitHub, Hugging Face, or another host is recorded

Table 6: Largest measured donor gains missed by the tested selectors.

Target task (ID: instruction)	Best measured donor (ID: instruction)	Success: NoDonor → mixed	Historical average	Instruction similarity	Trajectory similarity	Multimodal	+ calibration
14: put the wine bottle on top of the cabinet	23: pick up the BBQ sauce and place it in the basket 31: pick up the black bowl in the top drawer	56.7% → 83.3% (+26.7)	NoDonor (0.0)	NoDonor (0.0)	NoDonor (0.0)	Donor 33 (−30.0)	Donor 33 (−30.0)
16: turn on the stove	of the wooden cabinet and place it on the plate	73.3% → 93.3% (+20.0)	NoDonor (0.0)	NoDonor (0.0)	NoDonor (0.0)	NoDonor (0.0)	NoDonor (0.0)

Table 7: Physical layout of the v1.0.0 release artifacts.

Released path	Contents
data/tasks.jsonl	Forty task records with stable IDs, instructions, suite membership, calibration/eligibility summaries, and roster flag
data/episodes.jsonl	800 stable source references with task key, length, source hash, and roster flag; no demonstration bytes
data/training_runs.jsonl	588 complete run records with condition, target/donor, seed, checkpoint, episode pools, aggregate draws, and hashes
data/rollouts.jsonl.gz	5,880 run–scenario outcomes with success, executed steps, wall time, termination state, video hash, and exclusion flag
data/utilities.jsonl; data/utility_matrix.tsv	182 matched utility cells with success counts and denominator, plus the matrix view
benchmark/primary/features/*.npz and inventories	Episode and language arrays with extractor, source, shape, and device metadata
benchmark/primary/splits.json	Frozen outer and inner role-disjoint folds, action universes, and split audit fields
benchmark/primary/predictions/*.jsonl.gz	Per-method held-out target–donor predictions and record-level audit status
benchmark/completion/masks.jsonl.gz; benchmark/completion/trial_metrics.csv.gz	18,018 query/scoring masks and 180,180 method–budget result rows; summaries and audit are JSON/CSV

as provenance but is not accepted as a license grant. Generated media and model artifacts are reviewed separately because they may embed or derive from upstream demonstration episodes, simulator assets, or software. The reviewed inventory marks each class as included, reference-only, or excluded, and the publish-ready audit reports no remaining license blockers.

Governance material in v1.0.0 is limited to the Dataset and Benchmark Cards, `KNOWN_ISSUES.md`, attribution notices, and the component license inventory. The package does not

claim a named maintenance contact, support period, public leaderboard, changelog, or formal correction/takedown workflow.

Resource reporting is similarly bounded. The top-level manifests give per-file byte sizes; rollout records give per-scenario wall times; the feature inventory identifies one NVIDIA GeForce RTX 3090; and the completion audit reports 50.79 seconds wall time and 0.060 GiB peak RSS. Training-run records contain aggregate draw counts and artifact hashes but no training wall time, accelerator hours, peak GPU memory, or energy estimate. Version 1.0.0 therefore supports no aggregate training/rollout accelerator-hour or carbon claim.